

Step 1: Understand and Define the Problem (Analyse)

Following features must be included in the feeder:

1. Providing timely food for dogs and cats.
2. Record if the food has been consumed along with its amount.
3. Warn staffs about various issues.

The inputs and outputs required are:

INPUTS

1. Feeding times
Record the time when the dogs and cats are meant to be fed.
2. Time detector
Compare real time along with the recorded time of feeding to process the schedule.
3. Food amount sensor
Monitor the quantity of stored pet foods.
4. Bowl weight sensor
Monitor the quantity of food in the bowl to determine if the food has been consumed.

OUTPUTS

1. Alarm System
Activated alarm if there is any kind of problems in the system or if the food has not been eaten.
2. Dispenser
Provides pet food in the bowl.

Assumptions and Limitations

1. Weight System
Unable to determine if the food is consumed or thrown because it only measures the weight of the food.
2. No food classification
System is not capable of identifying if the dispenser contains pet food or other materials.

BLOCK DIAGRAM

